

Patent Document Report

Document Name: 50OR570US - Draft - QCed - 5-Aug-2026.docx

Date: 10 Sep 2026

Claims: 18 | Pages: 53 | Cancelled: 12, 20

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SUMMARY OF FINDINGS

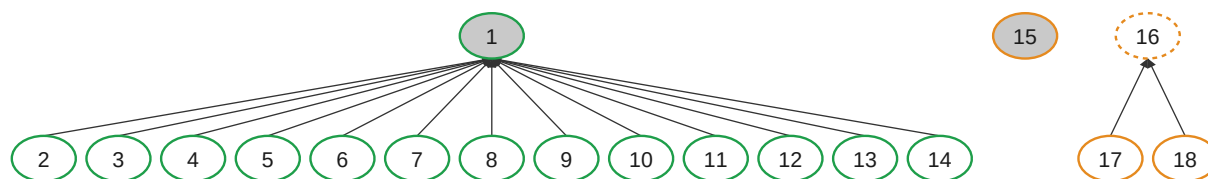
	Section	Result
I.	Claims Hierarchy	✗ 1 claim with an invalid parent reference
II.	Claim Errors and Warnings	✗ 1 error and 1 warning found
III.	Antecedents Errors/Warnings	✗ 22 errors and 12 warnings found
IV.	Claim Terms and Words Having No or Limited Support in the Specification	Not analysed
V.	Inconsistent Part Names and Numbers	Not analysed
VI.	Inconsistent Part Numbers in Figures	Not analysed
VII.	Inconsistent Acronyms	Not analysed
VIII.	Document Language Warnings	Not analysed

I. Claims Hierarchy¹

✗ 1 claim with an invalid parent reference.

- Claim 16 refers to itself.

The claim set contains **18** claims: **2** independent (1, 15) and **16** dependent.



Independent claim	Claim type	Dependent claims	Claims in tree
1	Method/process	2-14	14
15	Apparatus/device	None	1
16 invalid parent	Apparatus/device	17-18	3

Cancelled by amendment and not analysed: claims 12, 20.

Claim type color legend: **Method/process**; **Apparatus/device**; **Composition**; **Article of manufacture**; **112(6)**; **Product by process**; **Jepson**

Status indicator legend: **[O]**="Original"; **[PP]**="Previously Presented"; **[CA]**="Currently Amended"; **[N]**="New"; **[X]**="Cancelled"; **[W]**="Withdrawn"; **[WA]**="Withdrawn and Amended"; **[NE]**="Not Entered"

¹ Graphical claim trees only: the type of each claim is indicated by its colour. Independent claims are grey inside. Claims with invalid parent numbers are dotted. If present, status indicators will be provided in brackets next to the claim numbers.

II. Claim Errors and Warnings²

✗ 1 error and 1 warning found.

- Exemplary language: claim 14
- Claim depends on itself: claim 16

Claim	Severity	Error/Warning
14	Warning	Exemplary language: Exemplary language: "and/or". Examples in a claim leave it unclear whether the example limits the claim. Suggested fix: Remove the example or recite the limitation it is meant to cover. 35 U.S.C. 112(b); MPEP 2173.05(d)
16	Error	Claim depends on itself: Claim 16 refers to itself. Suggested fix: Point the dependency at the claim this one is meant to further limit. A claim renumbered by an amendment often keeps the cross-reference it had under the old numbering. 37 CFR 1.75(c)

² Fix suggestions/detailed explanations are not repeated for the same types of issues to preserve space.

III. Antecedents Errors/Warnings

✗ 22 errors and 12 warnings found.

- Missing antecedent basis: 21 in claims 14, 16-18
- Reverse antecedent: 1 in claim 17
- Limiting preamble: 8 in claims 1, 15
- Singular/plural mismatch: 4 in claims 4-5, 7-8

Claim	Error/Warning
1. A computer-implemented method^{1} for recovering a table structure^{2} from a global spectral representation^{3} of a document image^{4}, comprising: receiving the document image comprising a table; encoding the document image using a visual geometry encoder configured tofor generate generating a plurality of patch embeddings; predicting the global spectral representation of the document image using a spectral projection head configured to that projects the plurality of patch embeddings onto one or more harmonic bases defined by one or more frequency tags, wherein the global spectral representation corresponds to one or more complex waveforms, superposed using the one or more frequency tags, associated with one or more cells of the table; extracting, from the global spectral representation, a complex waveform of the one or more complex waveforms located within a frequency band isolated for a spectral coordinate on a frequency lattice; computing a cell energy associated with the spectral coordinate and determining one or more geometry coefficients from the complex waveform when the cell energy exceeds a threshold indicating an existence of a cell of the one or more cells at the spectral coordinate; reconstructing a cell boundary, based on a geometry mode count, corresponding to the cell of the one or more cells and sampling the cell boundary to generate one or more cell vertices, wherein the one or more cell vertices correspond to a polygonal boundary associated with the cell; and converting the one or more cell vertices into the polygonal boundary associated with the cell to recover a cell geometry of the table.	{1} Limiting preamble?: "a computer-implemented method." Terms in the preamble may be given patentable weight if re-used in the claim body. See, e.g., MPEP 2111.02. Suggested fix: Confirm this is intended to limit the claim. If it is only the intended use or field, consider whether the body should recite it; if it is a claim element, introduce it in the body instead. {2} Limiting preamble?: "a table structure." Terms in the preamble may be given patentable weight if re-used in the claim body. See, e.g., MPEP 2111.02. {3} Limiting preamble?: "a global spectral representation." Terms in the preamble may be given patentable weight if re-used in the claim body. See, e.g., MPEP 2111.02. {4} Limiting preamble?: "a document image." Terms in the preamble may be given patentable weight if re-used in the claim body. See, e.g., MPEP 2111.02.

Claim	Error/Warning
4. The computer-implemented method for recovering the table structure from the global spectral representation of the document image of claim 1, wherein extracting the complex waveform^{1} from the global spectral representation comprises extracting the complex waveform by applying an inverse Fourier projection to the global spectral representation based on the frequency band isolated on the frequency lattice.	{1} Singular/plural mismatch?: "the complex waveform." Found the same term in singular and plural forms. Suggested fix: Make the number consistent, or tie the reference to the set it comes from (for example "a suction conduit of the plurality of suction conduits").
5. The computer-implemented method for recovering the table structure from the global spectral representation of the document image of claim 1, wherein the frequency lattice is two-dimensional (2D) and is aligned with one or more rows and one or more columns of the table, and wherein a guard band enforces a spectral separation between one or more spectral coordinates^{1}, associated with the one or more cells, on the frequency lattice to prevent aliasing.	{1} Singular/plural mismatch?: "one or more spectral coordinates." Found the same term in singular and plural forms.
7. The computer-implemented method for recovering the table structure from the global spectral representation of the document image of claim 1, further comprises expressing the global spectral representation of the document image as a sum of one or more spectral components, wherein a first spectral component of the one or more spectral components corresponds to a table layout, a second spectral component of the one or more spectral components corresponds to one or more of a cell identity and a cell adjacency within the table based on one or more rows and one or more columns, and a third spectral component of the one or more spectral components corresponds to a plurality of polygonal boundaries^{1} associated with the one or more cells of the table, wherein the one or more spectral components occupy disjoint frequency bands.	{1} Singular/plural mismatch?: "a plurality of polygonal boundaries." Found the same term in singular and plural forms.
8. The computer-implemented method for recovering the table structure from the global spectral representation of the document image of claim 1, further comprises: annotating, during training, the cell of the one or more cells as a polygon and determining a parametric curve, defined by one or more two-dimensional coordinates, by traversing the polygonal boundary corresponding to the polygon using an arc-length normalization; generating the complex waveform associated with the cell by combining the one or more two-dimensional coordinates of the parametric curve;	{1} Singular/plural mismatch?: "a low-frequency basis." Found the same term in singular and plural forms.

Claim	Error/Warning
projecting the complex waveform onto a low-frequency basis^{1} based on the geometry mode count; and assigning a frequency tag of the one or more frequency tags to the complex waveform to superpose the complex waveform with the one or more complex waveforms to form a target spectral representation of the document image.	
14. The computer-implemented method for recovering the table structure from the global spectral representation of the document image of claim 1, wherein an ordering of one or more rows and/or one or more columns is induced by a frequency magnitude (Please double-check dependency of this claim, as none of the features^{1} claimed here is introduced in claim 1.).	{1} Missing antecedent basis: No antecedent basis for "the features" in claim 14. Suggested fix: Introduce the element first (e.g. "a feature"), or change "the features" to refer to an element that is already recited.
15. A system^{1} to recover a table structure^{2} from a global spectral representation^{3} of a document image^{4}, comprising: one or more processors; and one or more non-transitory computer-readable media storing instructions, which, when executed by the one or more processors, cause the one or more processors to perform a method comprising: receiving the document image comprising a table; encoding the document image using a visual geometry encoder configured to for generate generating a plurality of patch embeddings; predicting the global spectral representation of the document image using a spectral projection head configured tofor project that projectsing the plurality of patch embeddings onto one or more harmonic bases defined by one or more frequency tags, wherein the global spectral representation corresponds to one or more complex waveforms, superposed using the one or more frequency tags, associated with one or more cells of the table; extracting, from the global spectral representation, a complex waveform of the one or more complex waveforms located within a frequency band isolated for a spectral coordinate on a frequency lattice; computing a cell energy associated with the spectral coordinate and determining one or more geometry coefficients from the complex waveform when the	{1} Limiting preamble?: "a system." Terms in the preamble may be given patentable weight if re-used in the claim body. See, e.g., MPEP 2111.02. {2} Limiting preamble?: "a table structure." Terms in the preamble may be given patentable weight if re-used in the claim body. See, e.g., MPEP 2111.02. {3} Limiting preamble?: "a global spectral representation." Terms in the preamble may be given patentable weight if re-used in the claim body. See, e.g., MPEP 2111.02. {4} Limiting preamble?: "a document image." Terms in the preamble may be given patentable weight if re-used in the claim body. See, e.g., MPEP 2111.02.

Claim	Error/Warning
cell energy exceeds a threshold indicating an existence of a cell of the one or more cells at the spectral coordinate; reconstructing a cell boundary, corresponding to the cell of the one or more cells, by projecting the one or more geometry coefficients onto a harmonic basis function based on a geometry mode count and sampling the cell boundary to generate one or more cell vertices, wherein the one or more cell vertices correspond to a polygonal boundary associated with the cell; converting the one or more cell vertices into the polygonal boundary associated with the cell to recover a cell geometry of the table; and decoupling the cell geometry from a cell location using one or more layout anchors predicted in absolute image coordinates.	
16. The system^{1} to recover the table structure^{2} from the global spectral representation^{3} of the document image^{4} of claim 16, wherein the instructions^{5} further cause the one or more processors to determine the one or more geometry coefficients by: obtaining a discrete signal by sampling the complex waveform^{6} at one or more points spaced uniformly; determining a plurality of geometry coefficients by projecting the discrete signal onto a harmonic basis of the one or more harmonic bases; and retaining the one or more geometry coefficients of the plurality of geometry coefficients based on the geometry mode count^{7}, wherein the geometry mode count^{7} indicates a number of retained Fourier modes in the complex waveform^{6}.	<i>Claim 16 cannot inherit antecedent basis: its parent reference is invalid (see section II). The findings below follow from that, and most will resolve once the dependency is corrected.</i> {1} Missing antecedent basis: No antecedent basis for "The system" in claim 16. {2} Missing antecedent basis: No antecedent basis for "the table structure" in claim 16. {3} Missing antecedent basis: No antecedent basis for "the global spectral representation" in claim 16. {4} Missing antecedent basis: No antecedent basis for "the document image" in claim 16. {5} Missing antecedent basis: No antecedent basis for "the instructions" in claim 16. {6} Missing antecedent basis: No antecedent basis for "the complex waveform" in claim 16.

Claim	Error/Warning
	{7} Missing antecedent basis: No antecedent basis for "the geometry mode count" in claim 16.
17. The system^{1} to recover the table structure^{2} from the global spectral representation^{3} of the document image^{4} of claim 16, wherein the instructions^{5} further cause the one or more processors to isolate the frequency band^{6} by assigning a frequency center of the one or more frequency tags to the cell^{8} of the one or more cells on the frequency lattice^{7} and masking a global spectrum around the cell.	{1} Missing antecedent basis: No antecedent basis for "The system" in claim 17. {2} Missing antecedent basis: No antecedent basis for "the table structure" in claim 17. {3} Missing antecedent basis: No antecedent basis for "the global spectral representation" in claim 17. {4} Missing antecedent basis: No antecedent basis for "the document image" in claim 17. {5} Missing antecedent basis: No antecedent basis for "the instructions" in claim 17. {6} Missing antecedent basis: No antecedent basis for "the frequency band" in claim 17. {7} Missing antecedent basis: No antecedent basis for "the frequency lattice" in claim 17. {8} Reverse antecedent: "the cell" is referenced before "one or more cells" introduces it in claim 17. Suggested fix: Introduce "one or more cells" before it is referenced, or change the earlier "the cell" to an indefinite form and make the later mention the reference.
18. The system^{1} to recover the table structure^{2} from the global spectral representation^{3} of the document image^{4} of claim 16, wherein the geometry mode count^{5} determines a geometric resolution of the cell boundary^{6}.	{1} Missing antecedent basis: No antecedent basis for "The system" in claim 18. {2} Missing antecedent basis: No antecedent basis for "the table structure" in claim 18. {3} Missing antecedent basis: No antecedent basis for "the global spectral

Claim	Error/Warning
	representation" in claim 18. {4} Missing antecedent basis: No antecedent basis for "the document image" in claim 18. {5} Missing antecedent basis: No antecedent basis for "the geometry mode count" in claim 18. {6} Missing antecedent basis: No antecedent basis for "the cell boundary" in claim 18.

IV. **Claim Terms and Words Having No or Limited Support in the Specification**

This section analyses the specification and is not included in this report.

V. **Inconsistent Part Names and Numbers**

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